



Lesson None: Hacking Ethics

https://codehs.com/lms/assignment/206644998/lesson_plan

Description	In this lesson, students will explore and discuss the ethics and legality around hacking. A security hacker is someone who seeks to break through defenses and exploit weaknesses in a computer system or network. There are white hat hackers, who help companies find and protect exploits in their systems, and black hat hackers who hack maliciously.				
Objective	Students will be able to: • Identify the difference between white hat hacking and black hat hacking • Explain career opportunities in cybersecurity				
Activities	4.7.1 Video: Hacking Ethics & Legality 4.7.2 Quiz: Hacking Ethics & Legality 4.7.3 Connection: Penetration Testing 4.7.4 Free Response: Reflection: Penetration Testing 4.7.5 Connection: Hack the Pentagon? 4.7.6 Free Response: Reflection: Hack the Pentagon 4.7.7 Quiz: Ethical Hacker Agreement				
Prior Knowledge	No prior knowledge needed.				
Planning Notes	• Review the slides and activities for the lesson before the start of class • This lesson is an excellent opportunity to talk with students about cybersecurity employment opportunities. Consider having students research and present different cybersecurity career paths. • This is a shorter lesson and can be paired with a career research assignment or review class.				
Standards Addressed	<table><tr><th colspan="2">CSTA 2 Standards</th></tr><tr><th>Name</th><th>Description</th></tr></table>	CSTA 2 Standards		Name	Description
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Name	Description				

2-IC-23

Describe tradeoffs between allowing information to be public and keeping information private and secure.

CSTA MS Standards

Name	Description
MS-ALG-IM-10	Examine evidence of beneficial and harmful impacts, ethical issues, and biases of algorithms encountered in daily life.
MS-SYS-SE-33	Evaluate common types of cyber attacks and preventions.
MS-SOC-HU-43	Analyze how the decisions humans make when using computing technologies have ethical and social consequences.

CYBER ORG 6-8 Standards

Name	Description
6-8.DC.THRT	Describe various types of threat actors. Clarification statement: At this level, student discussions should focus on previous standards as well as grade-appropriate examples of various threat actors, such as nation-states, cyber terrorist groups, organized crime, or hacktivists.
6-8.DC.ETH	Distinguish between ethical and malicious hacking. Clarification statement: At this level, student discussions should focus on previous standards as well as grade-appropriate examples of the various types of hackers, occupations, and beliefs.

ISTE Standards

Name	Description
7b	Students use collaborative technologies to work with others, including peers, experts

or community members, to examine issues and problems from multiple viewpoints.

Teaching and Learning Strategies

Lesson Opener:

- Have students brainstorm and write down answers to the discussion questions listed below. Students can work individually or in groups/pairs. Have them share their responses. [5 mins]

Activities:

- Watch the lesson video and complete the corresponding quiz. [5-7 mins]
- Watch the *Penetration Testing* connection video and fill out the corresponding reflection. [10 mins]
- Explore the *Hack the Pentagon?* connection and fill out the corresponding reflection. [10 mins]
- Sign the *Ethical Hacker Agreement* by answering the quiz question. [2 mins]

Lesson Closer:

- Have students reflect and discuss their responses to the end of class discussion questions. [5 mins]

Discussion Questions

Beginning of Class:

- What is a “hacker”?
 - *A hacker is someone who seeks to break through defenses and exploit weaknesses in a computer system or network.*
- Are there any times when a hacker might be allowed to attempt to hack into a system?
 - *Yes. Hackers may be allowed to attempt to hack into a system to discover vulnerabilities that can then be fixed.*
- What are some cybersecurity career opportunities that you can think of?
 - *Find a whole list of careers [here!](#)*

End of Class:

- What is a white hat hacker?
 - *A white hat hacker is a computer security specialist who attempts to break into protected systems and networks to test and assess their security.*
- What is black hat hacking?
 - *Black hat hacking is attempting to find computer security vulnerabilities and exploiting them for personal financial gain or other malicious reasons.*
- What are the potential consequences of hacking illegally?
 - *For just hacking (not considering whether fraud or other illegal activity was committed as well) a person can face felony charges including prison time and high fines.*

Resources/Handouts

Vocabulary

Term		Definition
Modification: Advanced	Modification: Students Needing Additional Support	Modification: English Language Learners